

NAME: JAMES MWAURA

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Abstract

This study addresses the accelerating decline of the East African lion (*Panthera leo melanochaita*) in the Greater Mara ecosystem, where conventional monitoring methods have proven too slow and too fragmented to keep pace with emerging threats such as poaching, retaliatory killing, and habitat encroachment. Drawing on Design Science Research, the study proposes, builds, and evaluates a digital twin platform that brings together satellite telemetry, nightlight trend data, IoT sensor feeds, and machine learning-based threat detection into a single geospatial intelligence system for real-time ecosystem monitoring and predictive conservation management. The literature reviewed points to a clear gap: satellite remote sensing, machine learning classification, IoT sensor networks, and spatial database infrastructure have each been developed and applied on their own, but no existing system integrates them into a unified digital twin purpose-built for lion conservation at ecosystem scale. The platform will be implemented using a Waterfall development process and evaluated through a combination of functional, performance, and user acceptance testing involving conservation managers, rangers, and technical stakeholders from the Kenya Wildlife Service and the Mara Conservancy. The study is expected to contribute both a practical decision-support tool for conservation managers in the Greater Mara and a methodological contribution to the developing field of conservation informatics.

Declaration Page

Student's Declaration

This research proposal is my own original work and has not been submitted for the assessment of a diploma or degree in this or any other university.

Signature:  Date: Jul 29, 2026

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List of Acronyms

Acronym	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
AUC-ROC	Area Under the Curve – Receiver Operating Characteristic
DNB	Day/Night Band
DSR	Design Science Research
ERD	Entity-Relationship Diagram
ESA	European Space Agency
FAR	False Acceptance Rate
FRR	False Rejection Rate
GDAL	Geospatial Data Abstraction Library
GIS	Geographic Information System
GPS	Global Positioning System
IoT	Internet of Things
IT	Information Technology
IUCN	International Union for Conservation of Nature
KWS	Kenya Wildlife Service
ML	Machine Learning
OWASP	Open Web Application Security Project
PostGIS	Postgres spatial database extension (Geographic Information System for PostgreSQL)
REST/RESTful	Representational State Transfer
SDG	Sustainable Development Goal
SDLC	Software Development Life Cycle
SMART	Spatial Monitoring and Reporting Tool
SMS	Short Message Service
SPSS	Statistical Package for the Social Sciences
SUS	System Usability Scale
UAT	User Acceptance Testing
UNEP-WCMC	United Nations Environment Programme World Conservation Monitoring Centre
VIIRS	Visible Infrared Imaging Radiometer Suite
WDPA	World Database on Protected Areas
XGBoost	Extreme Gradient Boosting

CHAPTER ONE

INTRODUCTION

Over the past few decades, big cat populations have declined steadily worldwide, and the African lion has been especially hard hit. At the same time, advances in satellite technology — accelerated by the growing accessibility of commercial space launches — have made it possible to observe ecological trends across the globe in a level of detail that was previously out of reach. Paired with advances in computing power, these developments have opened the door to geospatial tools that are increasingly being put to work in conservation and habitat preservation.

1.1 Background to the Study

Wildlife conservation across sub-Saharan Africa is becoming more difficult as habitat fragmentation deepens and human-wildlife conflict grows more frequent. The East African lion (*Panthera leo melanochaita*) is one of the species most affected by this trend. Its population within the Greater Mara ecosystem — comprising the Maasai Mara National Reserve and the conservancies around it — has declined substantially over the past two decades.

The IUCN lists the African lion as Vulnerable, and the outlook in the Greater Mara has not improved: lions there face rising retaliatory killing, a shrinking prey base, and migration corridors increasingly disrupted by human activity (IUCN, 2021).

Traditional approaches to wildlife monitoring — periodic aerial surveys, ranger patrols, and handwritten field notes — are no longer able to keep pace with the volume of information conservation now requires. These methods tend to produce information that arrives late and covers only part of the picture, which makes a fast, well-targeted response difficult. Digital twin technology offers a way forward: a live, data-driven virtual replica of a physical environment that combines sensor feeds, mapping, and forecasting, and that has already shown its value in urban planning and environmental monitoring (Grieves & Vickers, 2017).

This study proposes a precision digital twin platform for the Greater Mara ecosystem, combining satellite telemetry, nightlight trend data, and AI-based threat detection into a single geospatial intelligence system. The intention is to give conservationists and researchers the ability to monitor the ecosystem in real time and anticipate emerging risks in a way that has not previously been possible.

1.2 Statement of the Problem

Lion populations in the Greater Mara are declining faster than in previous decades, and a central reason is the absence of real-time data analysis and reliable forecasting. Existing monitoring methods, such as field surveys, are costly and are consequently conducted infrequently. Other approaches — rangers spotting poaching activity, encroachment, or migration patterns as they occur — tend to be slow and inconsistent. At the same time, large volumes of data from sensors, satellite imagery, and incident reports go largely unused, since no single platform exists to bring them together and make sense of them. The result is that valuable information is wasted and conservation efforts fall short of what they could achieve (Pettorelli et al., 2018).

1.3 Objectives

1.3.1 General Objective

The general objective of this study is to design, develop, and evaluate a digital twin platform as a precision geospatial intelligence system for real-time monitoring, predictive modelling, and data-driven conservation management of the East African lion population in the Greater Mara ecosystem.

1.3.2 Specific Objectives

The study will be guided by the following specific objectives:

- i. To design a geospatial intelligence system integrating satellite telemetry and nightlight trend data for real-time monitoring of keystone species in the Greater Mara ecosystem.
- ii. To develop an AI-driven threat detection module capable of identifying human-wildlife conflict zones and habitat encroachment within the protected conservancy perimeter.
- iii. To develop and deploy the digital twin platform as a fully operational conservation intelligence system in the Greater Mara ecosystem.

1.4 Research Questions

This study will be guided by the following research questions:

- I. How can satellite telemetry and nightlight trend analysis be integrated into a unified geospatial digital twin for ecosystem monitoring in the Greater Mara?
- II. To what extent can AI-driven predictive modelling accurately detect human-wildlife conflict events within the Greater Mara conservancy?
- III. How can a digital twin be developed and deployed as a functional, user-facing conservation intelligence system that integrates geospatial visualisation and predictive modelling to support conservation in the Greater Mara ecosystem?

1.5 Hypothesis

H₁: The digital twin will improve the timeliness and accuracy of threat detection and ecosystem monitoring, compared with the conventional conservation monitoring methods currently used in the Greater Mara.

1.6 Significance of the Study

This study is expected to offer scientific, practical, and socio-environmental value. Scientifically, it is expected to contribute to the developing field of conservation informatics by demonstrating how digital twin technology can be applied to a complex, dynamic ecological system. Practically, the digital twin is intended to give wildlife managers, conservation rangers, and policy stakeholders in the Greater Mara an evidence-based, real-time decision-support tool capable of reducing response times to poaching incidents, improving patrol route planning, and supporting the management of human-wildlife conflict. Because the Greater Mara supports one of the largest lion populations on the continent, improvements in conservation management here could have implications well beyond the local area.

For communities living alongside the conservancy, more effective lion population management is expected to translate into fewer livestock losses from predation and better coexistence between people and wildlife. The study aligns with the UN Sustainable Development Goals — particularly SDG 15 (Life on Land) — and with Kenya's National Wildlife Conservation and Management Act (2013), contributing to national biodiversity targets and international conservation commitments.

1.7 Scope and Delimitation of the Study

This study focuses on the design and evaluation of a digital twin platform for the Greater Mara ecosystem in Kenya, encompassing the Maasai Mara National Reserve and its neighbouring conservancies. The primary focus is the East

African lion (*Panthera leo melanochaita*), although the geospatial tools developed are expected to allow for future expansion into multi-species monitoring.

Development and field testing of the platform will take place over a defined timeframe, drawing on satellite imagery from the VIIRS Day/Night Band and ESA WorldCover, together with LandDX terrain data and records from the World Database on Protected Areas.

1.8 Operational Definition of Key Terms

Digital Twin: a virtual computational replica of a physical system — in this context, the Greater Mara ecosystem — that integrates real-time data streams to mirror, simulate, and predict the behaviour of the physical environment (Grieves & Vickers, 2017).

Geospatial Intelligence: the extraction of actionable information from geospatial data, including satellite imagery, terrain maps, and spatial sensor outputs, applied here to ecological monitoring and conservation management.

XGBoost (Extreme Gradient Boosting): a scalable machine learning algorithm employing gradient-boosted decision trees, used in this study to develop predictive models for poaching activity detection and threat classification.

PostGIS: a spatial database extender for PostgreSQL that enables geographic object storage and geospatial query processing, serving as the primary spatial data infrastructure for the proposed digital twin platform.

Keystone Species: a species, such as the East African lion, whose presence and ecological role have a disproportionately large effect on its natural environment, and whose loss would fundamentally alter the ecosystem's structure.

WDPA: the World Database on Protected Areas, maintained by UNEP-WCMC, which provides globally standardised geospatial data on protected area boundaries and conservation designations.

Sentinel Analytics: real-time biological monitoring using satellite telemetry and on-ground acoustic sensors to continuously track species distribution and movement across spatial grid cells within the ecosystem.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

This chapter brings together the existing research most relevant to the development of a digital twin platform for lion conservation in the Greater Mara. Rather than covering every adjacent field, the review focuses on five areas that directly shape the design and rationale of this study: the current state of East African lion conservation; the origins and structure of digital twin technology; the use of geospatial tools and satellite data in wildlife monitoring; machine learning and AI-based threat detection; and technology-driven conservation efforts already underway in East Africa. The chapter closes by identifying the specific gap this study is designed to fill.

2.2 Conservation Status of the East African Lion

The East African lion (*Panthera leo melanochaita*) is listed as Vulnerable on the IUCN Red List. Across the African continent, lion numbers have dropped from an estimated 200,000 in the mid-twentieth century to under 25,000 today (IUCN, 2021). In the Greater Mara, this decline plays out through a familiar set of pressures: pastoralist communities killing lions in retaliation for livestock attacks, a shrinking prey base driven by bushmeat poaching and land degradation, and migratory corridors being steadily broken up by farms and settlements (Pettorelli et al., 2018).

The research record on this is consistent. Bauer et al. (2015) tracked a 42% drop in African lion populations over 21 years, linking it primarily to human-lion conflict and the inability of protected area management to respond fast enough to emerging threats. What comes through clearly in that body of work is that the problem is not simply the threats themselves — it is the lag between a threat appearing and a response being mounted. Closing that gap is precisely what this study sets out to do.

2.3 Digital Twin Technology: Theoretical Foundations

The digital twin concept was introduced by Grieves and Vickers (2017) in the context of industrial engineering. At its core, a digital twin is a virtual representation of a real-world system that stays synchronised with it through continuous data exchange. The idea started in manufacturing, but has since been applied to smart cities, supply chains, and increasingly to natural environments (Hämäläinen, 2020).

Its architecture rests on three elements:

1. The physical system being modelled.
2. An interactive digital representation.
3. A two-way data connection between them.

In a conservation context, the physical system is the ecosystem itself; the interactive representation is a computational model drawing on sensor feeds, satellite data, and historical records; and the connection is maintained through IoT networks and real-time data pipelines (Grieves & Vickers, 2017). This setup means that managers are not just watching what is happening, but can run scenarios, anticipate events, and test response options before committing to them in the field.

Hamalainen (2020) showed how this architecture scales when applied to complex, dynamic systems through smart city work, and the lessons transfer well to ecosystem management. The underlying challenge is similar: synthesising many layers of spatial data in real time and turning them into something decision-makers can actually use.

2.4 Geospatial Intelligence and Remote Sensing in Wildlife Conservation

Satellite-based monitoring has become a standard part of the conservation toolkit, and the quality of the data available has improved considerably as commercial satellite providers have entered the field. Datasets like ESA WorldCover and the VIIRS Day/Night Band give researchers continuous, wide-area coverage of land use change, vegetation health, and the spread of human settlements, all of which directly affect wildlife habitat (Pettorelli et al., 2018).

Pettorelli et al. (2018) reviewed how satellite remote sensing is being used in applied ecology and found that combining multiple data sources — vegetation indices, terrain data, and nighttime light imagery — produces a significantly richer picture of human-wildlife conflict risk in savanna environments than any single layer alone. That multi-source approach is directly reflected in the architecture of this study.

On the infrastructure side, PostGIS-enabled spatial databases have become the go-to solution for managing large volumes of conservation geospatial data. Their ability to handle complex spatial queries at scale makes them the right fit for a platform that needs to process data across hundreds of thousands of ecosystem grid cells in near real time (Hämäläinen, 2020).

2.5 Artificial Intelligence and Predictive Modelling for Conservation

Machine learning has steadily found its way into conservation over the past decade. XGBoost (Extreme Gradient Boosting) has emerged as one of the more reliable algorithms for ecological classification problems, including species distribution modelling, illegal activity detection, and habitat mapping, largely because it handles noisy, high-dimensional data well and is resistant to overfitting (Chen & Guestrin, 2016).

Nikkhah et al. (2021) showed how combining IoT sensor data with machine learning models can enable environmental monitoring systems to detect changes early enough to trigger a preventive response rather than a reactive one. That is a meaningful distinction in a conservation setting, where getting ahead of a poaching event or a conflict incident is far more valuable than recording it after the fact.

There is also growing evidence for the effectiveness of acoustic monitoring paired with AI classification. Studies have shown that sound-based detection systems can identify gunshot events in protected areas with accuracies above 85% (Pettorelli et al., 2018). That kind of performance forms part of the technical basis for the acoustic monitoring layer proposed in this study.

2.6 Technology-Based Conservation Frameworks in East Africa

East Africa has seen a number of technology-driven conservation efforts in recent years, which offer useful context for this study. The Kenya Wildlife Service has built GPS collar telemetry and aerial surveillance into its lion monitoring work, but these systems are largely reactive — they document what has happened rather than anticipating what might. Tools like SMART (Spatial Monitoring and Reporting Tool) have helped standardise how ranger patrol data is collected, but they stop short of providing the kind of real-time analytical feedback that would actually change how patrols are allocated on any given day (UNEP-WCMC, 2023).

Closer to home, Malenya (2025) demonstrated at Zetech University how sensor networks, IoT architecture, and data-driven analytics can be integrated into a functioning monitoring system in an agricultural context. The domain is different, but the methodological logic — fusing real-time sensor data to generate predictive insight — maps well onto what this study is attempting in wildlife conservation.

At the continental level, the African Wildlife Foundation's work with camera trap networks and ranger dashboards represents the current ceiling of what technology-enabled conservation looks like in Africa. But these remain largely separate systems, lacking the integrated geospatial intelligence and predictive capability that a properly realised digital twin would provide. That gap is what this study is directly responding to.

2.7 Research Gap

Looking across the literature, a clear gap emerges. The individual building blocks of the proposed platform — satellite remote sensing, machine learning threat detection, IoT sensor networks, and spatial database infrastructure — have each been developed and tested, but always in isolation or in fields adjacent to wildlife conservation. No study has yet brought these components together into a single, unified digital twin system purpose-built for lion conservation management at the scale of an ecosystem like the Greater Mara.

What exists in the region today is fragmented, reactive, and lacking in predictive power. The platform proposed in this study addresses that directly — not by adding another monitoring layer to an already crowded landscape, but by building the integrating architecture that has so far been missing. In doing so, it also opens a genuine contribution to the conservation informatics literature, where the application of digital twin methodology to African wildlife ecosystems remains largely unexplored.

CHAPTER THREE

RESEARCH METHODOLOGY

3.1 Introduction

This chapter will set out how the study is to be carried out. It will cover the research design, the overarching methodology guiding the work, the approach to be taken in building the system, the population the study will draw on, how participants will be sampled, the methods to be used to collect and analyse data, the tools to be relied on during development, and the ethical safeguards to be observed throughout. The project will follow Design Science Research (DSR) as its guiding methodology, since the aim is not simply to understand a problem but to design, build, and evaluate a working technological artefact — in this case, a digital twin platform for lion conservation. Once the design has been settled, the Waterfall model will carry the system through implementation in a structured, sequential manner. This combination reflects the study's dual purpose: understanding the conservation pressures facing East African lions, and building a functioning, predictive system capable of closing the gap in real-time monitoring and threat response identified in Chapter Two.

3.2 Research Design

The study will be built around Design Science Research (DSR) as its guiding design. This choice follows from what the study aims to achieve: designing, building, and evaluating a technological artefact that can meaningfully address human-wildlife conflict and poaching pressure in the Greater Mara Ecosystem.

DSR suits this kind of work because it is oriented towards solving practical problems by creating and testing information technology solutions (Hevner et al., 2004). Where purely observational research stops at describing a phenomenon, DSR pushes further, producing artefacts — systems, models, frameworks — that can be put to use in real settings to improve outcomes. The digital twin platform proposed here is expected to fit this description: rather than remaining a conceptual exercise, it will be a deployable system drawing together geospatial data, machine learning, and IoT sensor feeds into a single source of intelligence for conservation managers.

A few features of DSR line up with the aims of this study:

- **Problem-centric focus:** the study will start from a genuine problem — lion populations declining partly because threat responses arrive too late — which is the kind of issue DSR is built to tackle.
- **Artefact development:** the digital twin platform itself will be the artefact developed and studied.
- **Practical relevance:** the system is intended for eventual use by conservation agencies rather than remaining a purely academic exercise.
- **Iterative evaluation:** DSR builds in evaluation stages that will allow the artefact to be refined as weaknesses surface.

This approach is broadly consistent with how similar work in conservation informatics has tended to proceed, where developing new tools and platforms sits at the centre of both the academic contribution and the practical payoff (Pettorelli et al., 2018).

3.3 Design Science Research Methodology

The study will follow the six-phase Design Science Research process set out by Peffers et al. (2007), which balances structure with enough flexibility to accommodate a system of this complexity. Each phase is discussed below in relation to this project.

3.3.1 Problem Identification and Motivation

Chapter Two already established that the East African lion (*Panthera leo melanochaita*) sits in the Vulnerable category on the IUCN Red List (IUCN, 2021), with numbers falling steadily as a result of human-wildlife conflict, habitat fragmentation, and poaching (Bauer et al., 2015). The deeper issue is not simply that these pressures exist, but that too much time tends to pass between a threat appearing and a management response reaching it.

To build an evidence base around this problem, the study will draw on:

- **Literature review:** peer-reviewed research, conservation reports, and government documentation on lion populations and human-wildlife conflict.
- **Interviews:** conversations with conservation managers, rangers, and wildlife officers from the Kenya Wildlife Service (KWS) and the Mara Conservancy.
- **Document analysis:** incident reports, patrol logs, and conflict records maintained by conservation agencies.

Together, these sources are expected to help establish how serious the problem is and why a predictive, real-time monitoring platform is needed.

3.3.2 Define Objectives of the Solution

The second phase will set out the objectives the proposed digital twin platform is expected to meet. Drawing on the problem identified above and the surrounding literature, the study will work towards five objectives:

1. **Integrate multiple data sources:** bring satellite imagery and historical patrol records together into a single platform.
2. **Detect threats in real time:** process incoming data streams to flag poaching activity, such as gunshots, and emerging conflict hotspots.
3. **Forecast risk:** apply machine learning, specifically XGBoost, to predict which zones are likely to see poaching or conflict.
4. **Support scenario testing:** let conservation managers try out “what-if” scenarios before committing to a response strategy.
5. **Produce usable outputs:** generate risk heatmaps, real-time alerts, and concrete recommendations for deploying rangers.

These objectives map directly onto the gap identified in Chapter Two: the absence, in the Greater Mara, of an integrated system capable of predicting conservation threats rather than merely recording them after the fact.

3.3.3 Design and Development

This will be the most technically demanding phase, in which the platform's architecture and components will take shape. It will follow the Waterfall model described in Section 3.4, and will involve:

- **Database design:** building a PostgreSQL database with PostGIS extensions to manage spatial data across the Greater Mara.
- **Data pipeline architecture:** designing pipelines that can ingest satellite feeds, acoustic sensor data, and GPS telemetry in near real time.
- **Machine learning module:** building XGBoost models for threat prediction and classification.
- **User interface design:** creating dashboards that let conservation managers view risk maps, receive alerts, and run simulations.

By the end of this phase, a working prototype of the digital twin platform is expected to exist.

3.3.4 Demonstration

The artefact will be put in front of a simulated or real conservation setting to assess what it can do. In practice, this will mean using the system to:

- Process satellite imagery and sensor data drawn from the Greater Mara.
- Flag potential poaching events, for example, through acoustic gunshot detection.
- Generate heatmaps showing where human-wildlife conflict is likely.
- Produce alerts and deployment recommendations for rangers.

The demonstration is intended to show, in concrete terms, how the system addresses the conservation problem and supports the decisions conservation managers have to make. It will be carried out with input from stakeholders such as the Kenya Wildlife Service and the Mara Conservancy, so that the exercise stays grounded in practical relevance.

3.3.5 Evaluation

The evaluation phase will assess how well the platform performs, across several dimensions, summarised in Table 3.1.

Table 3.1: Evaluation Criteria

Evaluation Criteria	Description	Method
Detection Accuracy	Accuracy of threat detection (poaching events, conflict hotspots)	Comparison with ground-truth data (patrol reports, confirmed incidents)
Predictive Performance	Accuracy of XGBoost predictions	Cross-validation, confusion matrix, AUC-ROC
System Usability	Ease of use for conservation managers	User acceptance testing, System Usability Scale (SUS)
Response Time	Time from data ingestion to alert generation	Performance testing, latency measurements
System Reliability	System uptime, data integrity, error rates	Monitoring and logging during demonstration phase
User Satisfaction	Stakeholder satisfaction with system outputs	Surveys and interviews with end-users

Testing activities will include functional testing, performance testing, user acceptance testing, and expert evaluation. The results obtained are expected to indicate whether the system meets its intended objectives and requirements.

3.3.6 Communication

The final phase will focus on sharing what has been learned. Findings — the system itself, the evaluation results, and the recommendations arising from them — will be documented through:

- **The final project report:** a full academic account of the research process.
- **Peer-reviewed publications:** submitted to journals in conservation informatics and wildlife management.
- **Stakeholder presentations:** delivered to conservation agencies, including the Kenya Wildlife Service and the Mara Conservancy.
- **User and technical manuals:** prepared to support adoption and ongoing maintenance of the system.

This spread of outputs is intended to help the work feed back into both academic knowledge and practical conservation effort.

3.4 System Development Methodology

Implementation of the digital twin platform will follow the Waterfall Software Development Life Cycle (SDLC). This model has been selected because it moves through a fixed sequence of phases, each completed before the next begins — an approach that supports clear documentation, traceability, and manageable project oversight, all of which matter for a system this data-intensive and complex.

Waterfall suits this project for a few reasons:

1. **Clear requirements:** the requirements — data integration, ML-based prediction, real-time alerting — are already well defined.
2. **Structured development:** a sequential approach means each component will be fully designed before implementation begins.
3. **Documentation:** the model's emphasis on documentation is expected to support future maintenance and enhancement.
4. **Risk management:** working in stages should help surface problems earlier, lowering the risk of the project running off course.

3.4.1 Requirements Analysis

This phase will establish, and document, what the digital twin platform needs to do, both functionally and otherwise. Information will be drawn from:

- **Conservation managers:** to understand how decisions get made and what operational needs look like.
- **Rangers and field staff:** to understand how data is actually collected and where the friction is.
- **Technical stakeholders:** to define performance, security, and scalability requirements.

Functional Requirements

- Ingesting data from satellite sources (ESA WorldCover, VIIRS), acoustic sensors, and GPS collars.
- Managing and analysing spatial data through PostGIS.
- Predicting threats using an XGBoost-based machine learning model.
- Generating real-time alerts via SMS, email, and dashboard notifications.
- Running scenario simulations.
- Authenticating users and enforcing role-based access control.

Non-Functional Requirements

- A target system response time of under 5 seconds for data processing and alert generation.
- A target system availability of 99.5% uptime.
- Scalability to process data across hundreds of thousands of ecosystem grid cells.
- Security measures covering data encryption, secure APIs, and compliance with Kenya's Data Protection Act (2019).
- Dashboards intuitive enough for non-technical users.

This phase will conclude with a comprehensive requirements specification document.

3.4.2 System Design

The design phase will focus on the overall architecture of the digital twin platform.

3.4.2.1 System Architecture Design

The platform will be organised into a layered architecture:

- **Data layer:** a PostgreSQL/PostGIS database handling both spatial and non-spatial data.
- **Processing layer:** Python-based pipelines for ingesting, cleaning, and transforming data.
- **Machine learning layer:** XGBoost models handling prediction and classification.
- **Application layer:** web dashboards and APIs mediating user interaction.
- **Presentation layer:** visualisation tools rendering risk maps, alerts, and reports.

3.4.2.2 Database Design

An Entity-Relationship Diagram (ERD) will guide the database design, modelling:

- **Spatial data:** land cover, vegetation indices, and human settlement layers.
- **Sensor data:** acoustic detections, GPS collar positions, and weather readings.
- **Incident data:** conflict reports, poaching events, and patrol logs.
- **User data:** system accounts, roles, and permissions.

3.4.2.3 Machine Learning Model Design

The XGBoost model will be designed to do three things:

- **Classify threat levels:** sort grid cells into low, medium, or high risk.
- **Predict conflict zones:** flag areas where human-lion conflict is likely.
- **Detect poaching events:** distinguish gunshot from non-gunshot acoustic signals.

3.4.3 Implementation

Implementation will turn the approved design into working code. Table 3.2 lists the technologies to be used, along with the reasoning behind each choice.

Table 3.2: Development Technologies

Component	Technology	Rationale
Database	PostgreSQL with PostGIS	Industry-standard spatial database with robust query performance
Backend	Python (Django/Flask)	Extensive libraries for data processing and ML (Pandas, Scikit-learn, XGBoost)
Machine Learning	XGBoost, Scikit-learn	High performance for classification tasks; handles noisy data well
Frontend	HTML, CSS, JavaScript (React/Vue.js)	Responsive dashboards for non-technical users
Visualisation	Leaflet/Mapbox, Plotly, Chart.js	Interactive maps and charts
Data Pipelines	Apache Kafka or RabbitMQ	Real-time data streaming
APIs	RESTful APIs	Integration with external data sources
Version Control	Git/GitHub	Code management and collaboration

Over the course of this phase, six modules will be built and integrated:

- **Data ingestion module:** will collect and process satellite, sensor, and historical data.
- **Spatial analysis module:** will manage and query spatial data through PostGIS.
- **Machine learning module:** will train and deploy the XGBoost threat prediction models.

- **Alert generation module:** will send real-time notifications to conservation managers.
- **Dashboard module:** will visualise risk maps, alerts, and reports.
- **User management module:** will handle authentication and role-based access control.

3.4.4 Testing

The testing phase will check whether the developed system behaves as the specified requirements demand. Testing will run across several levels, summarised in Table 3.3.

Table 3.3: Testing Levels

Testing Level	Description	Tools/Methods
Unit Testing	Testing individual modules, such as data ingestion and ML prediction	PyTest, JUnit
Integration Testing	Testing how system components interact with one another	Postman, API testing
System Testing	Testing the complete application against the full set of requirements	Manual and automated testing
Machine Learning Testing	Evaluating model accuracy (FAR, FRR, AUC-ROC)	Cross-validation, confusion matrix
Performance Testing	Evaluating system response time and scalability under load	JMeter, load testing
User Acceptance Testing (UAT)	Testing with conservation managers and rangers	User feedback, SUS surveys
Security Testing	Testing for vulnerabilities and gaps in data protection	OWASP guidelines, vulnerability scanning

Testing is intended to catch and correct errors before deployment, so that what reaches users is reliable and performs as intended.

3.4.5 Deployment

Deployment will install the system in its target environment and configure everything it needs to run. This will involve:

1. **Server setup:** configuring cloud or on-premise servers with the required software (Python, PostgreSQL, PostGIS, and so on).
2. **Database initialisation:** creating database schemas and loading the initial spatial datasets.
3. **Model deployment:** putting the trained XGBoost models into production.
4. **API configuration:** setting up RESTful APIs for data ingestion and user interaction.
5. **User account creation:** setting up accounts for conservation managers and rangers.
6. **Training:** walking conservation managers and rangers through how to use the platform.
7. **Documentation:** preparing user manuals and technical documentation.

3.4.6 Maintenance

Maintenance activities will continue after deployment, to keep the system effective over time:

- **Corrective maintenance:** fixing errors and bugs identified during system use.
- **Adaptive maintenance:** updating the system to accommodate new data sources, models, or user requirements.

- **Preventive maintenance:** regular backups, performance optimisation, and vulnerability patching.
- **Enhancement:** adding new features, such as additional data sources or improved ML models.
- **User support:** providing ongoing support to address operational challenges.

A maintenance plan will be developed to help ensure the platform's long-term sustainability.

3.5 Target Population

The target population for this study will comprise stakeholders involved in lion conservation and wildlife management in the Greater Mara Ecosystem. These stakeholders are considered appropriate respondents because they interact directly with conservation systems and the decisions built on top of them. The target population will include:

1. **Conservation Managers and Administrators:** personnel responsible for strategic decisions regarding wildlife protection, ranger deployment, and resource allocation, drawn from the Kenya Wildlife Service (KWS), the Mara Conservancy, and other conservation organisations operating in the Greater Mara.
2. **Rangers and Field Staff:** individuals who conduct patrols, monitor wildlife, and record incidents in the field, and who are expected to supply ground-truth data on lion populations, human-wildlife conflict, and poaching events.
3. **Technical Personnel:** IT staff and data analysts working with conservation technologies, including GPS collars, camera traps, and existing monitoring tools such as SMART.
4. **Community Representatives:** members of pastoralist communities living adjacent to the Greater Mara who experience human-lion conflict directly and can offer insight into local perspectives and challenges.
5. **Academics and Researchers:** experts in wildlife ecology, conservation informatics, and related fields who can offer scholarly perspectives on the proposed platform.

3.6 Sampling Techniques and Sample Size

The study will employ purposive sampling to select respondents with relevant knowledge and experience of lion conservation, human-wildlife conflict, and conservation technologies in the Greater Mara. Purposive sampling is considered appropriate here because the study needs input from individuals with specific expertise rather than a broadly representative population sample.

3.6.1 Sampling Criteria

Respondents will be selected on the basis of the following criteria:

- Direct involvement in lion conservation or wildlife management in the Greater Mara.
- Familiarity with existing conservation technologies and monitoring systems.
- Willingness to participate and provide informed consent.
- Ability to contribute relevant insight to the system's design and evaluation.

3.6.2 Sample Composition

The sample will consist of approximately 30–40 respondents, distributed across the categories shown in Table 3.4.

Table 3.4: Sample Composition

Category	Number	Rationale
Conservation Managers	5–8	Strategic decision-makers, system requirements
Rangers and Field Staff	10–15	End-users, ground-truth data, usability testing

Category	Number	Rationale
Technical Personnel	3–5	Technical requirements, system integration
Community Representatives	5–8	Community perspectives, conflict dynamics
Academic Experts	3–5	Scholarly validation, expert evaluation

This sample size is considered appropriate for a qualitative and design-oriented study, where the focus is on depth of insight rather than statistical generalisability. The respondents selected are expected to offer insight into existing conservation challenges, user requirements, and feedback on the system once developed.

3.7 Data Collection Methods

Data collection will draw on several methods, so that findings can be triangulated and the research objectives covered from more than one angle. The methods below will be employed, though they are not expected to be equally productive in practice. Rangers and community members in the Greater Mara are often dispersed and difficult to reach with written instruments, and questionnaire response rates in field settings of this kind tend to be low. For this reason, the literature review and document analysis are likely to provide a more dependable base of evidence than the questionnaire, which will be used mainly to supplement, rather than substitute for, the interview- and document-based methods.

3.7.1 Interviews

Semi-structured interviews will be conducted with conservation managers, rangers, technical personnel, and community representatives, covering:

- Current conservation challenges and the threats facing lion populations.
- Experience with existing monitoring systems, such as SMART and GPS telemetry.
- Requirements for a digital twin platform.
- Perceptions of predictive analytics and real-time monitoring.
- The practical challenges of implementing new technology in a conservation context.

Tools: interview guides, audio recorders (with consent), and note-taking.

3.7.2 Questionnaires

Structured questionnaires will be administered to a wider group of conservation personnel, where feasible, gathering quantitative data on:

- Frequency and types of human-wildlife conflict incidents.
- Existing technology usage and satisfaction.
- Perceived gaps in current monitoring systems.
- Usability preferences for the proposed platform.

Given the practical difficulty of reaching dispersed field staff and community members with written instruments, response rates for this method may be modest. Findings from the questionnaire will therefore be treated as indicative rather than conclusive, and read alongside the interview and document-based evidence rather than in isolation.

Tools: paper-based or digital questionnaires (Google Forms).

3.7.3 Document Analysis

Existing documents and records will be reviewed to gather historical and contextual data, including:

- Incident reports and patrol logs from KWS and the Mara Conservancy.
- Wildlife census and lion population data.
- Existing conservation policies and strategies.
- SMART database records and reports.

Because these records already exist and do not depend on respondent participation, document analysis is likely to be one of the more reliable sources of evidence available to the study.

Tools: document review checklists, data extraction forms.

3.7.4 Observation

Direct observation will be conducted during ranger patrols and conservation operations to build an understanding of:

- Current data collection procedures.
- Workflows and decision-making processes.
- Challenges and constraints encountered in the field.
- Human-wildlife conflict dynamics.

Tools: observation checklists, field notes, photography (with relevant permissions).

3.7.5 System Logs and Sensor Data

During the demonstration and evaluation phases, system logs and sensor data will be collected automatically from:

- Acoustic sensors used for gunshot detection.
- Satellite data feeds covering land cover, vegetation, and nighttime lights.
- System interaction logs from the user dashboards.

This data will be used to help evaluate how well the system performs.

3.8 Data Analysis

Data analysis will combine quantitative and qualitative techniques, reflecting the mixed-methods nature of the study.

3.8.1 Qualitative Data Analysis

Qualitative data from interviews, observations, and open-ended questionnaire responses will be analysed using thematic analysis:

1. **Familiarisation:** reading through all the qualitative material to build an overall understanding.
2. **Coding:** identifying and labelling key concepts, ideas, and recurring themes.
3. **Theme development:** grouping related codes into broader themes.
4. **Interpretation:** reading those themes against the research objectives and the literature.

Tools: NVivo, or manual coding using spreadsheets.

3.8.2 Quantitative Data Analysis

Quantitative data from questionnaires, system logs, and sensor data will be analysed using descriptive and inferential statistical techniques:

- **Descriptive statistics:** frequencies, percentages, means, and standard deviations to summarise survey responses.
- **Machine learning evaluation metrics:** accuracy, precision, recall, F1-score, confusion matrix, and AUC-ROC to evaluate the XGBoost model.
- **Performance metrics:** system response time, detection latency, and uptime.

Tools: Python (Pandas, Scikit-learn), Excel, SPSS.

3.8.3 Spatial Data Analysis

Spatial data from satellite imagery and geographic information systems will be processed and analysed using:

- **PostGIS:** spatial queries and data management.
- **QGIS:** visualisation and spatial analysis.
- **Geospatial libraries:** GeoPandas, Rasterio, and GDAL in Python.

3.8.4 Presentation of Findings

The analysed data will be presented using:

- **Tables:** for summarising quantitative data.
- **Charts and graphs:** for visualising survey results and performance metrics.
- **Maps:** for presenting spatial data and risk heatmaps.
- **Narrative descriptions:** for discussing qualitative themes and interpretations.

3.9 Ethical Considerations

The study will adhere to established ethical principles throughout the research process. The following considerations will guide the research.

3.9.1 Informed Consent

Participation in the study will be entirely voluntary. Every respondent will receive a clear explanation of the study's purpose, procedures, and their rights before data collection begins, and written or verbal informed consent will be obtained from all participants.

3.9.2 Confidentiality and Anonymity

Confidentiality of participant information will be maintained. Identifiable information will be stored separately from research data, pseudonyms will be used where possible, and data will be reported in aggregate form to help prevent identification of individual respondents.

3.9.3 Data Protection and Storage

All collected data will be stored securely, with access restricted to the researcher. Data will be kept on password-protected computers and encrypted cloud services. Biometric and sensitive data will be anonymised and securely deleted after analysis, in compliance with Kenya's Data Protection Act (2019).

3.9.4 Privacy and Non-Interference

The study will respect the privacy of participants and communities. Field observations will be conducted discreetly, and photography will only be undertaken with explicit permission. The study will aim to avoid interfering with ongoing conservation operations.

3.9.5 Academic Integrity

The research will be conducted with honesty and transparency. All sources will be properly cited, and findings will be reported accurately, without fabrication or misrepresentation.

3.10 Summary

This chapter has set out the research methodology to be followed in developing a digital twin platform for lion conservation in the Greater Mara Ecosystem. The study will adopt a Design Science Research approach, providing a structured framework for designing, developing, and evaluating the digital twin as a technological artefact. The Waterfall Software Development Life Cycle model will carry the implementation forward, providing a systematic and sequential process from requirements analysis through to maintenance.

The target population will comprise conservation managers, rangers, technical personnel, community representatives, and academic experts, selected through purposive sampling. Data collection will combine interviews, document analysis, observation, and system logs, supplemented by questionnaires where feasible, with data analysed through thematic analysis, descriptive statistics, and machine learning evaluation metrics. Ethical considerations, including informed consent, confidentiality, and data protection, will be adhered to throughout the research process.

Taken together, this methodological framework is intended to support the development of a functional digital twin platform that addresses the identified conservation gap, while contributing to both academic knowledge and practical conservation efforts.

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Appendices

Appendix 1: Letter of Introduction

Dear Sir/Madam,

RE: REQUEST FOR PARTICIPATION IN A RESEARCH STUDY

My name is James Mwaura, a student at Zetech University pursuing a Diploma in Software Engineering under the Department of Information Communication and Technology. I am carrying out a research study titled "*A Digital Twin Platform for East African Lion Conservation in the Greater Mara Ecosystem*" as a requirement for the completion of my diploma.

The purpose of this study is to design, develop, and evaluate a digital twin platform to support real-time monitoring and predictive conservation management of the East African lion population in the Greater Mara. I am kindly requesting your participation in this study, either through an interview, a questionnaire, or by providing relevant documentation, as applicable.

Any information you provide will be treated with strict confidentiality and used solely for academic purposes. Participation is entirely voluntary, and you are free to withdraw at any time without consequence.

Thank you in advance for your time and cooperation.

Yours faithfully,

James
DSE-02-0143/2025
Department of Information Communication and Technology,

Zetech University

Mwaura

Appendix 2: Research Instruments

A. Interview Guide (Conservation Managers, Rangers, and Technical Personnel)

1. What are the most significant challenges you currently face in monitoring lion populations and human-wildlife conflict in the Greater Mara?
2. What monitoring tools or systems (e.g. SMART, GPS collars) do you currently rely on, and what are their main limitations?
3. How much time typically passes between a threat (e.g. poaching, conflict incident) occurring and a response being mounted?
4. What would a real-time, predictive monitoring system need to do to be genuinely useful to your work?
5. What concerns, if any, would you have about adopting a new digital monitoring platform?
6. What features would you consider essential in a dashboard or alert system designed for ranger deployment decisions?

B. Questionnaire (Wider Conservation Personnel)

1. How frequently do you encounter human-wildlife conflict incidents in your area of operation? (Daily / Weekly / Monthly / Rarely)
2. Which existing technologies do you currently use for monitoring wildlife or patrols? (Please list)
3. How satisfied are you with these existing tools? (Very satisfied / Satisfied / Neutral / Dissatisfied / Very dissatisfied)

4. What gaps, if any, exist in current monitoring systems?
5. How comfortable are you using digital dashboards or mobile applications for fieldwork? (Very comfortable / Comfortable / Neutral / Uncomfortable / Very uncomfortable)
6. What features would you most want to see in a new conservation monitoring platform?

Appendix 3: List of Organisations in the Population

- Kenya Wildlife Service (KWS)
- Mara Conservancy
- Narok County Government, Department of Tourism and Wildlife
- Local pastoralist community associations adjacent to the Greater Mara
- Zetech University, Department of Information Communication and Technology